

Exercise 15A

Random Variables and Frequency Tables

FOUNDATION

- 1 State whether each random variable is numeric or categorical. If it is numeric, state whether it is discrete or continuous. Comments may be appropriate.
- a The favourite day of the week for a person chosen at random in Australia
 - b Height of Australian professional basketball players
 - c Age
 - d Political affiliation
 - e Colour of a counter drawn from a cup containing 5 red and 6 blue counters.
 - f Sex (male or female) of a child attending a particular primary school
 - g Sum of the numbers when two dice are thrown
 - h Shoe size
 - i Examination scores

- 2 Each experiment below returns a numerical result. Define a random variable X for each experiment and write down domain of possible values that X can take. This is also called the *sample space* for the experiment.
- a A pupil is investigating the length of words chosen at random from the sentence *The quick brown fox jumped over the lazy dog*.
 - b Four die are tossed to investigate the number of sixes that are turned up.
 - c A reseacher is carrying out a census of a street in a city to determine household size.
 - d A digit is chosen at random from the 12-digit number 141421356237.

- 3 Cards in a set each have one of the numbers 1–5 written on them at random. The cards are shuffled. A card is selected and the number is recorded. This experiment is repeated; at each stage the card is returned to the pack and the cards are reshuffled. The frequency of each outcome is shown in the table below. Let the random variable X be the number written on a card selected at random.

x	0	1	2	3	4	5
f	10	20	30	25	5	10

- a Record the relative frequencies in a new row in the table.
- b Estimate:
 - i $P(X = 1)$
 - ii $P(2 \leq X \leq 4)$
 - iii $P(1 \leq X < 4)$
 - iv $P(X = 6)$
 - v $P(X \leq 2)$
 - vi $P(X < 4)$
 - vii $P(X \geq 1)$
 - viii $P(X > 1)$
 - ix $P(X \text{ is even})$

- 4 In a seaside town, researchers have asked random people how many days they have visited the local beach in the last week. Their responses are aggregated in the table below.

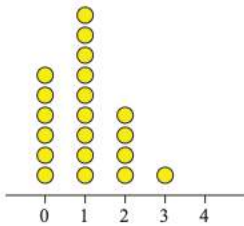
x	0	1	2	3	4	5	6	7
f	3	6	4	2	1	1	3	5
rf	0.12	0.24	0.16	0.08	0.04	0.04	0.12	0.20

- a Express each of these probabilities in the form $P(X = x)$ or $P(X \leq x)$. The probability that a random person from the town visits the beach:
 - i four times a week
 - ii less than or equal to four times a week
 - iii no more than twice a week
 - iv less than four times a week
 - v more than twice a weak
 - vi between 3 and 4 times a week (inclusive).
- b Estimate each probability from part a using the relative frequencies from the table.

5 A die is tossed 50 times and the results recorded in the following table.

x	1	2	3	4	5	6
f	7	5	6	7	8	17
rf						

- a Complete the table by filling in the row of relative frequencies.
 - b Is the die fair?
 - c What is the most likely outcome? What name is given to this outcome?
 - d Estimate the probability of throwing a 4.
 - e Estimate the probability of throwing a number bigger than 3.
 - f Estimate the number of sixes obtained if the dice is thrown 800 times.
 - g Find the probability that the sum of two dice rolls is 11.
- 6 The number of siblings for each member of a class is recorded in the dot plot below.



- a Fill in a table showing the frequency and relative frequency as a decimal.
 - b What is the modal number of siblings?
 - c What is the percentage probability that a student chosen at random has no more than 2 siblings?
 - d What is the percentage probability that a family chosen at random from the families represented in the class has only a single child?
- DEVELOPMENT**
- 7 A teacher sets a simple pre-unit quiz before starting a new topic. The quiz is out of 10. The results are in the following table.

x	1	2	3	4	5	6	7	8	9
f	1	0	0	2	4	5	3	1	2

- a Draw a simple histogram showing this data. Your histogram bars should centre around the scores 1, 2, 3, ... 9 and there should be no gaps between the bars. Your graph should include a title and a label on each axis.
- b Join the centres of neighbouring bars to form a frequency polygon. By convention, the polygon is joined back to the horizontal axis at 0 and 10.
- c Comment on the shape of the distribution of the data.
- d If you have access to a spreadsheet program, you should be able to draw this graph as a column (bar) graph with the gaps between the bars set to 0. By adding a second graph to the same axes, you can superimpose the frequency polygon.

8 Clive is a frequent visitor to the zoo, and he is always interested to see how many lions are out of their dens and visible to the public. He records the number of lions he sees on each visit. The relative frequency of his results are shown in the following table.

x	1	2	3	4	5	6
rf	0.4	0	0.16	0	0.12	0.32

- a** Draw a simple relative frequency histogram showing this data. A vertical unit of 0.1 unit as 1 cm is recommended. Your histogram bars should centre around the scores 1, 2, 3, . . . , 6 and there should be no gaps between the bars. Your graph should include a title and a label on each axis.
 - b** Join the centres of neighbouring bars to form a relative frequency polygon. By convention, the polygon is joined back to the horizontal axis at 0 and 7.
 - c** Can you suggest a story to explain any patterns you see in the data? It may need further investigation to check your explanation.
 - d** If you have access to a spreadsheet program, you should be able to draw this graph as a column (bar) graph with the gaps between the bars set to 0. By adding a second graph to the same axes, you can draw the frequency polygon.
- 9** A school is investigating the number of times students buy lunch at the canteen in a week. They sample a group of students at random. Their experimental data is in the table below.

x	0	1	2	3	4	5
f	29	13	3	0	1	4

- a** Discuss what the school needs to consider when designing this experiment.
 - b** How many students are sampled?
 - c** What do you think X represents?
 - d** Add a relative frequency row.
 - e** What is the probability that a student chosen at random buys their lunch sometime during the week?
 - f** What is the probability that a students buys their lunch every school day?
 - g** The school population has 1200 students. Estimate how many lunches the canteen sells during the week.
- 10** Amy scoops up coins at random from a pile containing one 10 c coin and two 5 c coins.
- a** Name the 10 c coin T and the two 5 c coins F1 and F2, and write down as sets the seven possible non-empty scoops.
 - b** Let the random variable X be the value of the money that she picks up. What values can X take?
 - c** Amy repeats the experiment 20 times, replacing the coins after each selection. Complete the following table of her results, including the relative frequency.

x	5	10	15	20
f	4	8	6	2
rf				

- d** Assuming that she is equally likely to pick up any of these seven non-empty scoops, draw up a probability tree diagram for X and calculate the theoretical probability of each outcome, adding your calculations to the table in a row marked p .
- e** Compare the theoretical probabilities with Amy's experimental outcomes.

11 An experiment with two possible outcomes is called a *Bernoulli* experiment. The outcome might be *Yes, No; True, False; In, Out; Success, Fail; Correct, Incorrect*, to give a few examples. If we take these two outcomes as 0 or 1, then a Bernoulli experiment generates a discrete random variable. A dart player records 1 if he hits the bulls eye, and 0 otherwise. His results are recorded in the following table.

x	0	1
f	40	7

- a** Add a relative frequency column.
 - b** Explain what the frequency and relative frequency mean in this case.
 - c** How many bull's eyes would he expect in 20 throws?
- 12** A teacher tells their class that the probability a pair of numbers chosen at random have no factors in common is $\frac{6}{\pi^2}$. A member of the computing and informatics club uses the GCD (*Greatest Common Divisor*) function in Excel to check if randomly selected pairs of numbers less than 100 have factors in common. Their results are displayed below.

x	0	1
f	31	19

- a** Discuss the design of their experiment.
 - b** Estimate the probability that a pair of numbers will have no factors in common. Does the experiment agree with the teacher's claimed theoretical probability?
 - c** Write your own spreadsheet to test this.
- 13** Most scientific calculators have a button to generate a random number. Your calculator may accept a function like `RandInt#(1,6)` or `RandBetween(1,6)`, generating a number between 1 and 6 inclusive. Your calculator may have a function like `Ran#` generating a decimal between 0 and 1. Typing `Ran#×6+1` will generate a random dice roll; just discard any decimal portion remaining.
- a** Use your calculator to simulate rolling a dice 50 times. Record your results in a table with a tally column.
 - b** Calculate the frequency and relative frequency of each outcome.
 - c** Is the dice simulated by the calculator weighted towards any particular result?
 - d** If each outcome is equally likely, we call the distribution of probabilities *uniform*. Is the distribution you obtained uniform?
 - e** This simulation exercise can also be carried out in a spreadsheet.

ENRICHMENT

- 14**
- a** Simulate the sum of the outcomes when a pair of dice is rolled on your calculator. As you roll the dice, record your results in a dot plot. Roll a pair of die 50 times.
 - b** Compare the results obtained by simulation in a spreadsheet. You can easily toss the pair of dice 500 times, for a more accurate simulation.

Exercise 15B

Cumulative Frequency

FOUNDATION

- 1 Find the median, mode and range of each dataset.
- a** 10 13 14 14 15 17 18
- b** 5 7 8 9 10 12 13 15 17
- c** 3 3 4 5 7 9 10 12 13 15
- d** 4 4 4 6 6 6 7 7 8 8 9 10
- e** 4 2 6 4 7 3 4 6 3 4 3 5 2 1 5 7 8
- f** 2 9 7 6 4 3 2 7 8 9 10 5 4 2 3 6 9 3

- 2 Construct a table of frequencies and cumulative frequencies for the following data. Find the number of scores, the mode(s) and the median.
- a** 2 2 2 3 3 4 4 4 4 5 5 6 6 8 8 9
- b** 1 4 7 6 3 8 9 3 4 6 6 8 4 3 6 1 2 2 5
- c** 9 3 3 5 4 9 12 10 6 12 10 14 7 7 3 10 12 7 4 8
- d** 5 4 7 3 3 6 7 7 7 6 7 6 2 3 7 7 7 7 7 3 3

- 3 The results of an experiment to investigate the value of a random variable X are listed in the following table.

x	0	1	2	3	4	5	6
f	3	2	4	2	5	3	2

- a** Copy the table and add a row listing cumulative frequencies and cumulative relative frequencies (to 3 decimal places).
- b** Estimate
- i** $P(X \leq 2)$ **ii** $P(X \leq 5)$ **iii** $P(2 < X \leq 5)$
- c** Estimate the median by finding x such that $P(X \leq x) = 0.5$.
- 4 The results of an experiment to investigate the value of a random variable X are listed in the following table.

x	0	1	2	3	4	5	6
f	2	4	6	12	8	5	1

- a** Copy the table and add a row listing cumulative frequencies and relative cumulative frequencies.
- b** Estimate
- i** $P(X \leq 1)$ **ii** $P(X \leq 5)$ **iii** $P(1 < X \leq 5)$
- c** Estimate
- i** $P(X \leq 3)$ **ii** $P(X > 3)$

- 5 Copy and complete the missing rows in these tables.

a	x	0	1	2	3	4	5	6
	f							
	cf	1	6	9	17	21	28	30

b	x	0	1	2	3	4	5	6
	f							
	cf	8	12	15	16	26	34	39

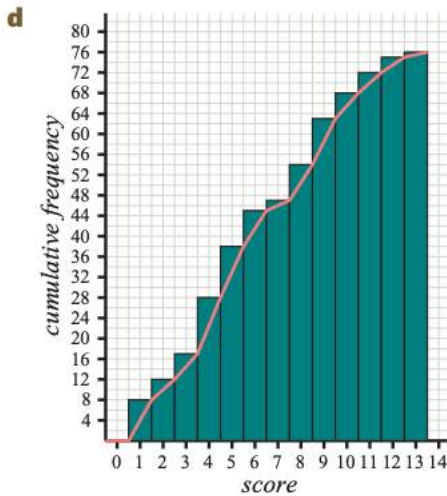
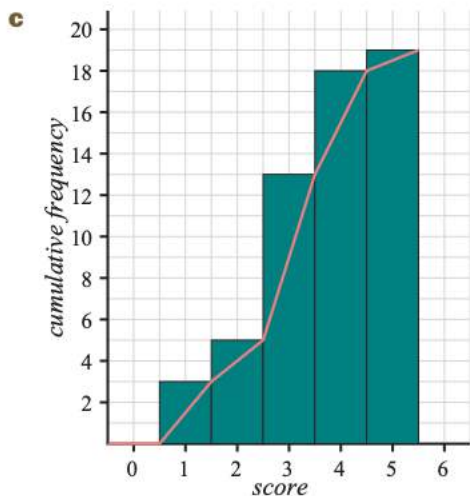
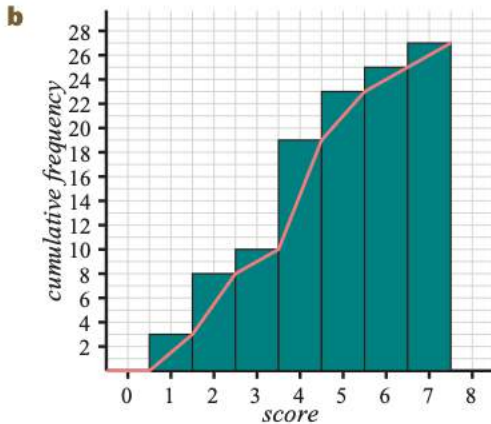
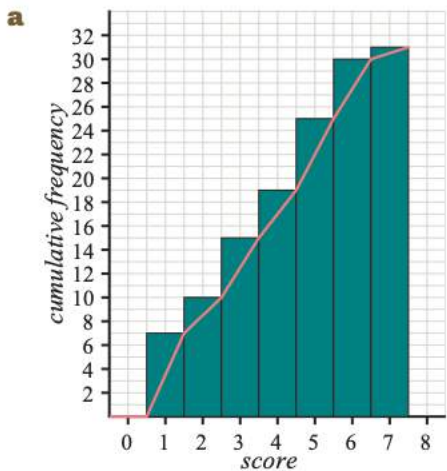
c (An experiment with 40 scores)

x	0	1	2	3	4
cf					
rf	0.15	0.225	0.05	0.225	0.35

d

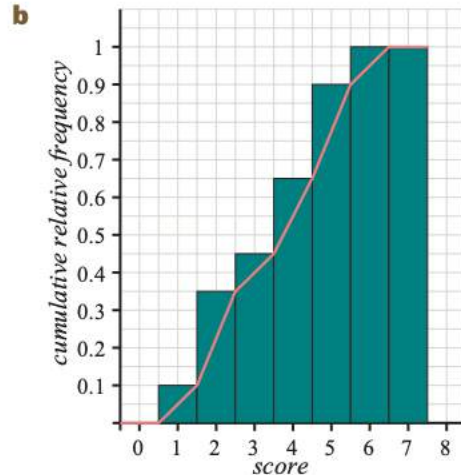
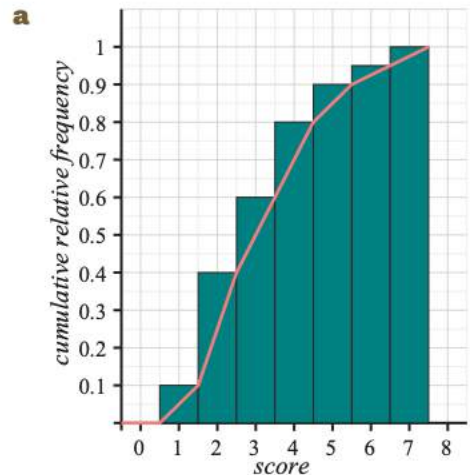
x	2	3	4	5	6
f					
cf					25
crf	0.2	0.52	0.60	0.88	1

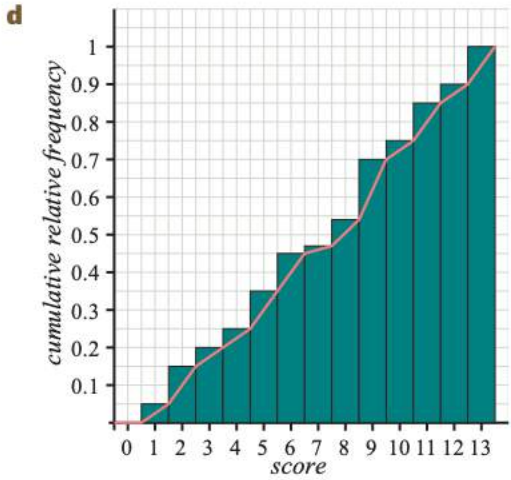
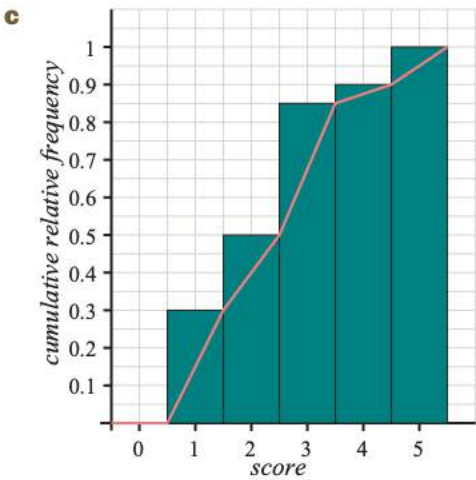
6 Use the cumulative frequency polygon and cumulative frequency histogram to find the number of scores and the median for each experiment.



DEVELOPMENT

7 Use the cumulative relative frequency polygon and cumulative relative frequency histogram to estimate the median for each experiment.





8 A teacher set a simple pre-unit quiz out of 10. The results are repeated again in the following table.

<i>x</i>	1	2	3	4	5	6	7	8	9
<i>f</i>	1	0	0	2	4	5	3	1	2

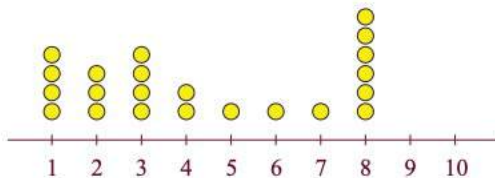
- a** Copy the table and add a cumulative frequency row.
- b** Draw a simple cumulative frequency histogram showing this data. Your histogram bars should centre around the scores 1, 2, 3, ... 9 and there should be no gaps between the bars. Your graph should include a title and a label on each axis.
- c** Join the **endpoints** each bar to its next neighbour to form a cumulative frequency polygon. By convention, the polygon is joined back to the horizontal axis at the start of the first bar.
- d** How many students got no more than 50%?
- e** How many students got less than 50%?
- f** The teacher is interested in what score 75% of the students failed to attain. What was the score?
- g** If you have access to a spreadsheet program, you should be able to draw this graph as a column (bar) graph with the gaps between the bars set to 0. By adding a second graph (a line graph joining the class centres) to the same axes, you can superimpose the frequency polygon.

9 As was mentioned in the previous exercise, Clive is a frequent visitor to the zoo. The random variable *X* is the number of lion sightings on each visit, and the relative frequency of his outcomes are shown in the following table.

<i>x</i>	1	2	3	4	5	6
rf	0.4	0	0.16	0	0.12	0.32

- a** Copy the table and add a cumulative relative frequency row.
- b** Draw a simple cumulative relative frequency histogram showing this data. A vertical unit of 0.1 unit as 1 cm is recommended. Your histogram bars should centre around the scores 1, 2, 3, ... 9 and there should be no gaps between the bars. Your graph should include a title and a label on each axis.
- c** Join the **endpoints** each bar to its next neighbour to form a cumulative relative frequency polygon. By convention, the polygon is joined back to the horizontal axis at the start of the first bar.
- d** If you have access to a spreadsheet program, you should be able to draw this graph as a column (bar) graph with the gaps between the bars set to 0. By adding a second graph to the same axes, you can superimpose the frequency polygon.

- 8.



- b** Construct a table of cumulative frequencies, and draw a cumulative bar chart showing the number of rides sold up to and including that day of the week.

13 An operator tracks the number of customers who pay the \$4 fee to take his amusement ride on each day of the week. His data is displayed in the following table.

Mon	Tue	Wed	Thu	Fri	Sat	Sun
13	32	35	38	57	75	65

- a** Draw a column chart showing the data, with days of the week on the horizontal axis. Use a scale of 1 cm per 10 rides.
- b** Construct a table of cumulative frequencies, and draw a cumulative bar chart showing the number of rides sold up to and including that day of the week.

ENRICHMENT

14 The median is sometimes called the *second quartile*, since 50% (two quarters) of the data is no bigger than the median. Twenty-five percent of the data is less than or equal to the first quartile, and seventy-five percent of the data is less than or equal to the third quartile.

- a** Estimate the first, second and third quartile for the data displayed in the histograms in Question **7a** and **7c**. Assume the data is discrete.
- b** Construct box plots for the two distributions. (You should have met box plots in your studies in previous years. You may have to do some research if you have forgotten these graphs.)

15 Sometime during the last year, a large restaurant began to hire out its five separate dining spaces for special group functions on Friday or Saturday night. The data collected has been used to generate the following cumulative relative frequency table. The random variable X is the number of rooms booked out on a night.

x	0	1	2	3	4	5
crrf	0.15	0.275	0.4625	0.6625	0.85	1

Record keeping was poor, and the manager is not sure when they started hiring out the rooms and how many nights are represented in the data.

- a** Find the mode, that is the most common number of rooms booked out on a night.
- b** Find the number of nights that the modal number of rooms were booked.

Exercise 15C

Grouped Data

FOUNDATION

- 1 A horticulturalist has planted 40 dwarf apple trees and he is investigating the yield of fruit produced per tree when the trees are 3 years old. The yield, in kilograms, is given in the following data set.

18 24.5 30.2 27.8 50.4 34.4 53.8 47.5 64.4 57.2
45.8 43.2 50.5 62.3 27.9 34.7 38.9 46.1 34.1 38.5
43.6 38.4 39 46.7 48 52.3 57.1 43.4 46.4 38.9
41.2 43.7 53.4 47 44.5 38.9 35 36.2 43.2 45.7

- a Use a tally table to group the data set into classes of 5: [15, 20), [20, 25), [25, 30), ...

lower class bound	15	20	25	30	35	40	45	50	55	60
tally										
frequency										

- b Hence construct a frequency histogram for the grouped data.
- c Add the frequency polygon to your graph.
- d i If spreadsheet software is available, use it to construct the histogram. You will need to set the bin width to 5. To ensure the first class begins at 15, you may need to choose the underflow bin to 20 (which means scores ≤ 20 form the first class).
- ii In your spreadsheet, adjust the bin width and decide on the best class size to display the data.
- e Add a cumulative frequency and cumulative relative frequency row to your table.
- f Construct a cumulative relative frequency histogram for your grouped data.
- g Use your cumulative relative frequency histogram to estimate the median of the grouped data. Do this by estimating, to the nearest kilogram, the weight of a fictitious tree at the 50% mark on your cumulative relative frequency polygon.
- h i Identify which classes represent trees with yield in the lower 20%.
- ii It is suggested that trees falling in the lower 20% should be removed. Discuss.

- 2 A teacher records the Examination results for their class:

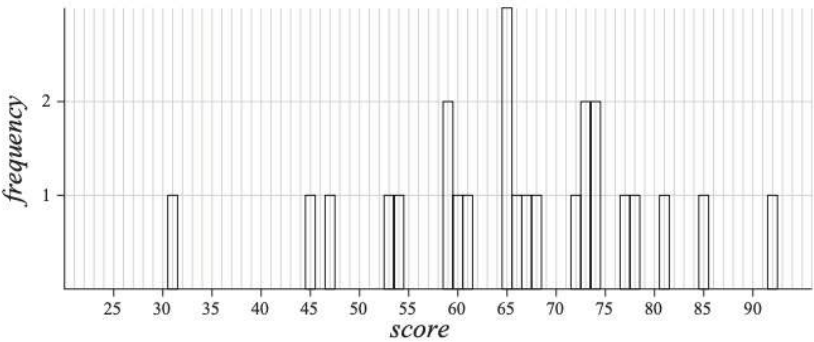
47, 85, 65, 73, 31, 66, 45, 65, 74, 81, 92, 74, 77, 65, 54, 53, 67, 78, 72, 61, 60, 59, 68, 59, 73

- a Copy the table below and fill in the missing values to group the data in bins of width 10 as shown.

x	[25, 35)	[35, 45)	[45, 55)	[55, 65)	[65, 75)	[75, 85)	[85, 95)
class centre							
Tally							
frequency							

- b Construct a frequency histogram for the grouped data. Add a frequency polygon.
- c The teacher also draws up a histogram with bin width 1, as below.

Exam Scores



Discuss the advantages and disadvantages of the grouped data histogram you displayed, compared with the teacher's histogram.

- d Draw up a table and construct a histogram with bins of width 5, that is [30, 35), [35, 40), ...

3 A business provides its employees with a mobile phone. Phones were purchased at the start of January and replaced over a three year period, due to faulty hardware, damage, planned upgrade or request for new features. The relative frequency for the replacement time period for a phone in months is shown in the following table. The data is grouped in 6 months bins. Thus, for example, 10% of phones are replaced between 12 and 18 months. The random variable X represents the lifespan of a random mobile phone in the company, before it is replaced.

x	[0, 6)	[6, 12)	[12, 18)	[18, 24)	[24, 30)	[30, 36]
rf	0.2	0.05	0.1	0.27	0.15	0.37

- a** X is a continuous random variable. Explain why $P(X = 6) = 0$ and write down the value of $P(X = x)$ for any x .
- b** Given $P(X < 6) = 0.2$, what is $P(X \leq 6)$?
- c** Estimate:
- i** the probability a phone is replaced in the first year, $P(X < 12)$;
 - ii** the probability a phone is replaced in the second year, $P(12 \leq X < 24)$;
 - iii** the probability a phone is replaced within the first 6 months of a year.
- 4** The business in the previous question has a similar scheme for laptops, but it replaces them over a 2 year period. The cumulative relative frequency for the replacement time period in months is shown in the following table. The data is grouped in 4 months bins. The random variable X represents the lifespan of a random laptop in the company, before it is replaced.

x	[0, 4)	[4, 8)	[8, 12)	[12, 16)	[16, 20)	[20, 24]
crf	0.12	0.22	0.37	0.57	0.79	1

- a** What fraction of laptops are replaced in the first year?
- b** Estimate the probability that a laptop lasts more than 8 months.
- c** Estimate:
- i** $P(4 < X < 20)$
 - ii** $P(X > 16)$.

DEVELOPMENT

5 Xiomi recorded her time to get to work each day. Her results in minutes were:

22 30 23.5 27 25 21.5 39 30 32.5 33 35.5 37 42 22 23.5 27 29.5 23 34

- a** Write the data out in order, and determine the median.
- b** Group the data into classes by completing the following table.
- | | | | | | | |
|--------------|-------|-------|-------|-------|-------|-------|
| class | 20–24 | 24–28 | 28–32 | 32–36 | 36–40 | 40–44 |
| class centre | | | | | | |
| frequency | | | | | | |
| cumulative | | | | | | |
- c** Find the median of the grouped data. Does it agree with your answer to part **a**?
- d** Draw a frequency histogram and polygon on the same chart.
- e** Draw a cumulative frequency histogram and polygon on the same chart.

6 In a class experiment, students measured their heights. The results in centimetres were:

155 152 165 162 170 168 165 162 166 154 158 159

163 166 164 164 159 157 163 154 166 158 159 163

- a** Display the data in a frequency table.
- b** Calculate the median of the dataset.
- c** Why would it not be helpful to graph the data without first grouping it into classes?

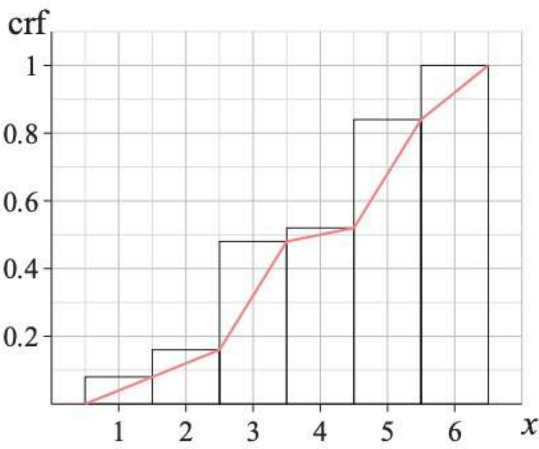
- d** Group the data into the intervals 150–154, 154–158, 158–162, 162–166, 166–170 and display your results in a grouped frequency table. Include any scores on a boundaries in the lower group, thus x is in the group 150–154 if $150 < x \leq 154$. (This is deliberately a different convention than we gave in the theory. Various conventions exist and you should always *read the question carefully*.)
- e** Calculate the median of the dataset from this grouped frequency table.
- f** Display your grouped data on a histogram with a frequency polygon joining the centres.
- g** Construct a cumulative frequency histogram and cumulative frequency polygon — remember that the polygon starts at the bottom left-corner of the first rectangle and passes through the right-hand top corner of each rectangle.
- h** Trace the line at frequency 12 (50%) until it meets the cumulative frequency polygon, and check whether this agrees with your answer for the median of the grouped data in part **e**.
- i** Construct a cumulative frequency histogram and cumulative frequency polygon of the *ungrouped* data.
- j** Compare your grouped and ungrouped cumulative histograms in parts **f** and **h**. How similar are the graphs? Contrast the differences between the histogram of the grouped data in part **f** and what you would expect the histogram of the ungrouped data to look like (not drawn).
- k** Confirm that the line at frequency 12 meets your ungrouped polygon to give the same median as in part **g**.
- l** If you have a spreadsheet program available, use it to construct a frequency histogram of the grouped data. The author found that to do this in Excel, he had to set the bin width to 4 and the underflow bin to 154 to force it to the desired starting bin. Excel follows the convention that data 154–158 is grouped as (154,158], as in this question. These distinctions become less important with continuous data and large data sets.

ENRICHMENT

7 Understanding the source and nature of your data is important, because it can affect how you interpret it and also how you display or manipulate the data set. Whether data is discrete or continuous is one example of this. Continuous data is always grouped, even if only because of rounding.

x	1	2	3	4	5	6
f	2	2	8	1	8	4
rf	0.08	0.08	0.32	0.04	0.32	0.16
crf	0.08	0.16	0.48	0.52	0.84	1

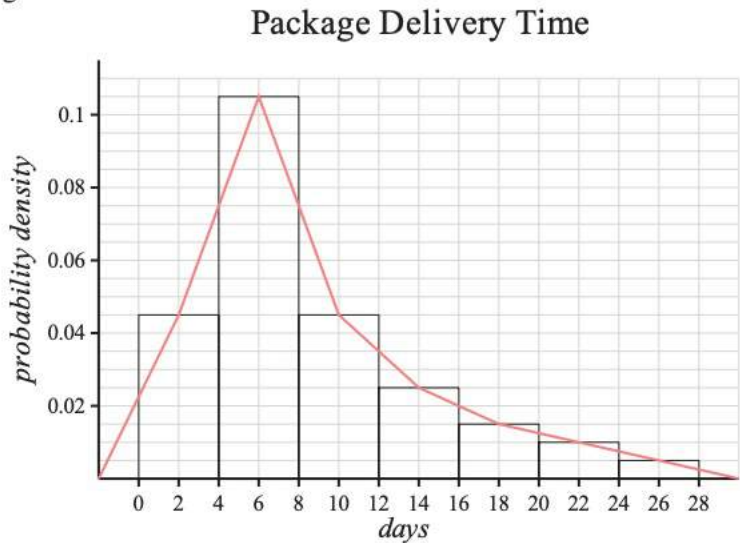
Consider the data above. A cumulative relative frequency histogram of the data is shown below.



Assume that this data is **discrete**.

- a** What is the range of the data (highest - lowest)?
- b** Use the cumulative frequency histogram to estimate the median (second quartile) and the upper and lower quartiles (that is, the scores at 50%, 25% and 75%).

- c Confirm your result by calculating the medians directly from the scores and frequencies in the table. (Recall that the lower quartile is defined to be the median of the data before the second quartile and the upper quartile is the median of the data after the second quartile).
- Assume now that the data is **continuous**. The scores represent class centres.
- d What is the length (range) of the interval with class centre 1?
 - e What is the range of the data (highest - lowest)?
 - f Use the cumulative frequency histogram to estimate the median and the upper and lower quartiles. Your estimates may not be whole numbers.
 - g A student looks at the table of cumulative relative frequencies, and gives the answer 3.5 for the median, because 0.5 is half way between 3 and 4. What is wrong with this argument?
 - h The upper quartile may be found without drawing the graph by finding the equation of the line between $P(4.5, 0.52)$ and $Q(5.5, 0.84)$ and the value of x when $y = 0.75$. What value does this give? Is it justified to give this apparently more accurate answer?
 - i What is the 40th percentile (also called the fourth decile) — that is, the score such that 40% of the scores lie on or below?
- 8 It is common for a histogram to plot *probability density* on the vertical axis, which is relative frequency divided by the bin width. The probability of an outcome in a class is the area of the rectangle (bin width $\times$ probability density).
- A delivery company tracks the time taken for packages to arrive at their destination. The data is shown on the following histogram.



- a Find the probability that the package will arrive in between 4 and 8 days.
- b Find the probability that the package will arrive in between 8 and 16 days. The polygon represents the smoothing out of the probability densities.
- c What is the total area under the polygonal function?
- d Estimate the probability that the package will take more than 18 days.

1a categorical

b numeric and continuous. But 'height correct to the nearest mm' is numeric and discrete.

c numeric and continuous. But 'age in years' is numeric and discrete.

d categorical by party or political code. This would need to be defined carefully — if a person can be affiliated to two parties, it would not be a function.

e categorical

f categorical

g numeric and discrete

h Shoe sizes are often arranged into categories.

i These are frequently integers from 1–100, that is, numeric and discrete. If results are reported by a grade, for example, A, B, C, ..., this might be considered categorical.

2a Let X be the number of letters in the chosen word $X \in \{3, 4, 5, 6\}$

b Let X be the number of sixes that are rolled. $X \in \{0, 1, 2, 3, 4\}$

c Let X be the number of members in the household. X is a positive integer less than or equal to the maximum household size (perhaps about 7? The research will need to decide if they are asking about number of children, size of immediate family or possible extended families with grandparents living onsite.)

d Let X be a digit chosen at random from the number 1.41421356237.

Then $X \in \{1, 2, 3, 4, 5, 6, 7\}$.

3a

x	0	1	2	3	4	5
f	10	20	30	25	5	10
rf	0.1	0.2	0.3	0.25	0.05	0.1

b i 0.2

ii 0.6

iii 0.75

iv 0

v 0.6

vi 0.85

vii 0.9

viii 0.7

ix 0.45

4a i $P(X = 4)$

ii $P(X \leq 4)$

iii $P(X \leq 2)$

iv $P(X \leq 3)$

v $P(X > 2) = 1 - P(X \leq 2)$

vi $P(X \leq 4) - P(X \leq 2)$

b i 0.04

ii 0.64

iii 0.52

iv 0.60

v 0.48

vi 0.12

5a

x	1	2	3	4	5	6
f	7	5	6	7	8	17
rf	0.14	0.10	0.12	0.14	0.16	0.34

b No - outcomes are not equally likely, and 6 is the most common outcome, which is frequently the desired outcome in dice games.

c 6. This is called the mode.

d 14%

e 64%

f $0.34 \times 800 = 272$

g Assuming independence, $0.16 \times 0.34 + 0.34 \times 0.16 = 0.1088$, i.e. 10.88%

6a

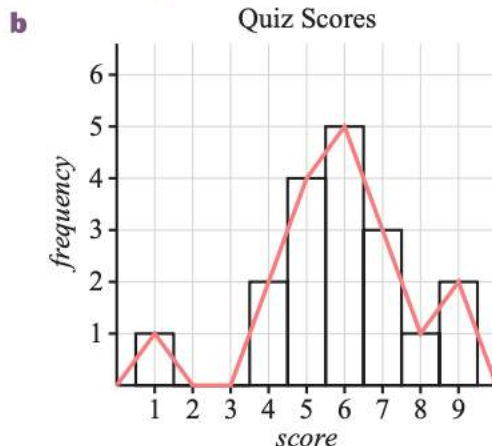
x	0	1	2	3
f	6	9	4	1
rf	0.30	0.45	0.2	0.05

b 1

c 95%

d 30%. Note we are assuming there are 20 families represented in the data, but the class may include children from the same family.

7a See part **b**

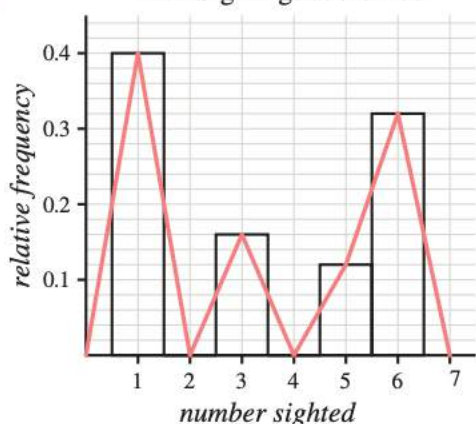


c We don't know how hard the quiz was, but the teacher should certainly take note of the student with a weak score of 1, and may need to do some extra work with the students on 4. Most of the

results are above 50%, which is as the teacher would want.

8b

Lion Sightings at the Zoo



- c There appears to be a single lion who is often out in all weathers enjoying the outdoors. There are also two lions who rarely come out on their own, but if one comes out the other does too. If the weather is good, perhaps around midday, most of the pride (5 lions) will come out. There is one less sociable lion who only comes out at feeding time. Investigating whether there is any commonality in what lions appear in the tabulated numbers (do they come out in groups), and whether it depends on weather or time of day, would test this explanation.

- 9b 50 Does anyone buy lunch twice in a day? Is this the data from a *random* week? The week should include a full five days and be otherwise unremarkable (middle of the term?). How is the student sample chosen? — they should not all be from the same year group, for example.

- c X should be the number of lunches a student buys in the week.

x	0	1	2	3	4	5
f	29	13	3	0	1	4
rf	0.58	0.26	0.06	0	0.02	0.08

- e 42%

- f 8%

- g They sell $1200 \times (0.26 + 2 \times 0.06 + 4 \times 0.02 + 5 \times 0.08) = 1032$ lunches

- 10a {T}, {F1}, {F2}, {T,F1}, {T,F2}, {F1,F2}, {T,F1,F2}

- b 5 c, 10 c, 15 c or 20 c

x	5	10	15	20
f	4	8	6	2
rf	0.20	0.40	0.30	0.10

x	5	10	15	20
p (frac)	$\frac{2}{7}$	$\frac{2}{7}$	$\frac{2}{7}$	$\frac{1}{7}$
p (decimal)	0.29	0.29	0.29	0.14

- e The results are not in close agreement, however Amy's sample size was very small so close agreement is unlikely. She should also look at how to make the process of drawing coins random.

x	0	1
f	40	7
rf	0.85	0.15

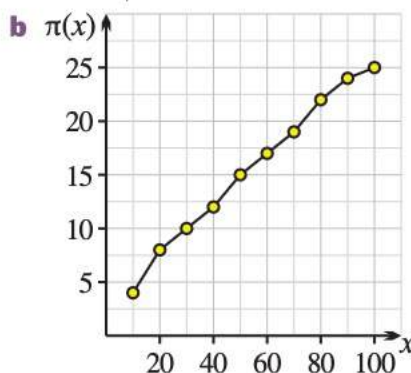
- b The number of bull's eyes and the fraction of bull's eyes (which is an estimate of how often he hits the bull's eye).

- c About 3

- 12a Obviously most numbers are not 2 digit, so the experiment assumes two digit numbers are representative of all numbers for this result.

- b $\frac{31}{50} \div 62\%$. The theoretical result is $6 \div \pi^2 \div 61\%$, so the agreement is quite good.

x	1	2	3	4	5	6	7	8	9	10
f	0	1	1	0	1	0	1	0	0	0
$\pi(x)$	0	1	2	2	3	3	4	4	4	4



- c There are $22 - 12 = 10$ primes: 41, 43, 47, 53, 59, 61, 67, 71, 73, 79

Exercise 15B

- 1a median 14, mode 14, range 8

- b median 10, every score is trivially a mode, range 12

- c median 8, mode 3, range 12

- d median 6.5, modes 4 & 6, range 6

- e median 4, mode 4, range 7

- f median 5.5, modes 2 & 3 & 9, range 8

2a

<i>x</i>	2	3	4	5	6	7	8	9
<i>f</i>	3	2	4	2	2	0	2	1
<i>cf</i>	3	5	9	11	13	13	15	16

16 scores; mode 4; median $(4 + 4)/2 = 4$

b

<i>x</i>	1	2	3	4	5	6	7	8	9
<i>f</i>	2	2	3	3	1	4	1	2	1
<i>cf</i>	2	4	7	10	11	15	16	18	19

19 scores; mode 6; median 4

c

<i>x</i>	3	4	5	6	7	8	9	10	11	12	13	14
<i>f</i>	3	2	1	1	3	1	2	3	0	3	0	1
<i>cf</i>	3	5	6	7	10	11	13	16	16	19	19	20

20 scores; modes 3, 7, 10, 11; median $(7 + 8)/2 = 7.5$

d

<i>x</i>	2	3	4	5	6	7
<i>f</i>	1	5	1	1	3	10
<i>cf</i>	1	6	7	8	11	21

21 scores; mode 7; median 6

3a

<i>x</i>	0	1	2	3	4	5	6
<i>f</i>	3	2	4	2	5	3	2
<i>cf</i>	3	5	9	11	16	19	21
<i>crf</i>	0.143	0.238	0.429	0.524	0.762	0.905	1

b i 0.429 ii 0.905 iii 0.476

c Because this is discrete data, we conclude that this will occur in the column defined by $x = 3$. If we were to treat this as continuous data, then we might interpolate the line through $(0.429, 2)$ and $(0.524, 3)$ with gradient $(0.524 - 0.429)/(3 - 2) = 0.095$ and equation $y - 0.429 = 0.095(x - 2)$, which gives $x = 2.75$ when $y = 0.5$.

4a

<i>x</i>	0	1	2	3	4	5	6
<i>f</i>	2	4	6	12	8	5	1
<i>cf</i>	2	6	12	24	32	37	38
<i>crf</i>	0.053	0.158	0.316	0.632	0.842	0.974	1

b i 0.211

ii 0.974

iii $0.974 - 0.158 = 0.816$

c i 0.632 ii $1 - 0.632 = 0.368$

5a

<i>x</i>	0	1	2	3	4	5	6
<i>f</i>	1	5	3	8	4	7	2
<i>cf</i>	1	6	9	17	21	28	30

b

<i>x</i>	0	1	2	3	4	5	6
<i>f</i>	8	4	3	1	10	8	5
<i>cf</i>	8	12	15	16	26	34	39

c

<i>x</i>	0	1	2	3	4
<i>cf</i>	6	15	17	26	40
<i>rf</i>	0.15	0.225	0.05	0.225	0.35

d

<i>x</i>	2	3	4	5	6
<i>f</i>	5	8	2	7	3
<i>cf</i>	5	13	15	22	25
<i>crf</i>	0.2	0.52	0.60	0.88	1

6a 31 scores; the median is the 16th score, viz 4

b 27 scores; the median is the 14th score, viz 4

c 19 scores; the median is the 10th score, viz 3

d 76 scores; the median is the average of the 38th and 39th score, viz 5.5

7a 3

b 4

c 2.5

d 8

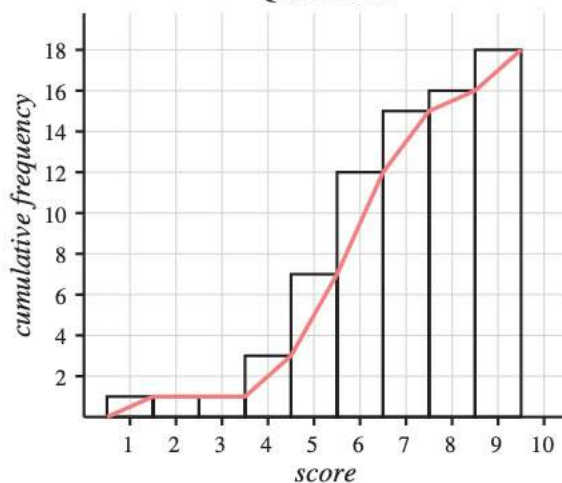
8a

<i>x</i>	1	2	3	4	5	6	7	8	9
<i>f</i>	1	0	0	2	4	5	3	1	2
<i>cf</i>	1	1	1	3	7	12	15	16	18

b See the next answer.

c

Quiz Scores



d 7

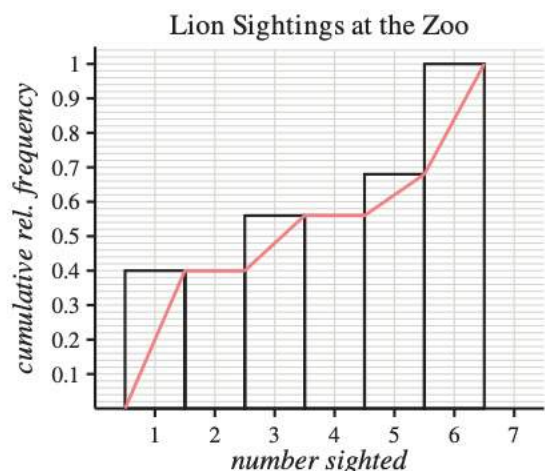
e 3

f The score that three-quarters of the students lie below, i.e. 13.5 students. Twelve students (67%) got at most 6, while fifteen students (83%) got at most 7.

9a

<i>x</i>	1	2	3	4	5	6
<i>rf</i>	0.4	0	0.16	0	0.12	0.32
<i>crf</i>	0.4	0.4	0.56	0.56	0.68	1

b See the next answer.

c**10a**

x	1	2	3	4	5	6	7	8
f	6	3	5	2	1	1	1	6
	4	7	11	13	14	15	16	22
crf	0.18	0.32	0.5	0.59	0.64	0.68	0.73	1

b 3.5**c** 8

d i This is a median, but it might be more useful to use the mode in this case. It may be easier to develop a square box for four cupcakes rather than three.

ii See the previous comments. It is also common for sales to package a larger box to encourage customers to overbuy.

iii This is the mode, but if a box of four is marketed, customers can just pick up two boxes of four.

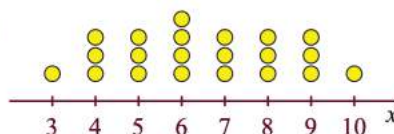
11a

x	1	2	3	4	5	6
cf	0	0	2	4	10	6
f	0	0	2	6	16	22
crf	0	0	0.09	0.36	0.73	1

d 20 students rank lower than him, hence he is placed 67% from the bottom. This suggests a band 5 result.

e Around 27% of his cohort obtain band 6, i.e. 8 students, so he needs to raise his rank about 2 places.

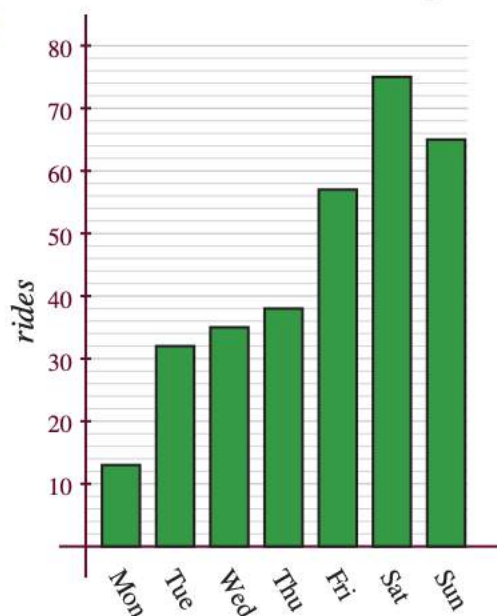
f There might be multiple students ranked 10th in his cohort; Cohorts vary in strength from year to year; Two places suggests a small change, but this is only after one assessment. He should be encouraged, but aim for a bigger increase to be sure.

12a

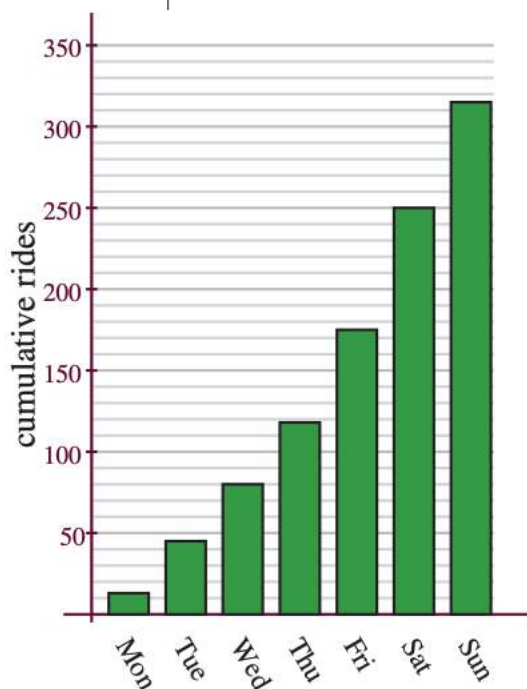
score x	3	4	5	6	7	8	9	10
frequency	1	3	3	4	3	3	3	1
cumulative	1	4	7	11	14	18	20	21

c 6 hoops**d** 6.5 hoops

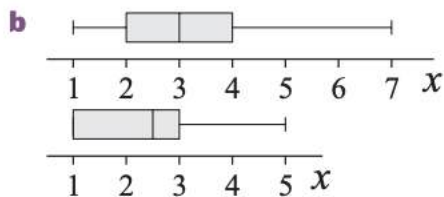
e Not really. If the scores are ordered by time, his scores improve over the sessions. This information is lost in the table and plot.

13a

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
frequency	13	32	35	38	57	75	65
cumulative	13	45	80	118	175	250	315



14a 1, 2, 3, 4, 7 and 1, 1, 2.5, 3, 5



15a The difference between the cumulative relative frequencies will generate the relative frequencies:

x	0	1	2	3	4	5
crf	0.15	0.275	0.4625	0.6625	0.85	1
rf	0.15	0.125	0.1875	0.2	0.1875	0.15

The mode is 3, since this is the most common outcome.

b Write the relative frequencies as fractions:

x	0	1	2	3	4	5
rf	$\frac{3}{20}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{5}$	$\frac{3}{16}$	$\frac{3}{20}$

It is now evident that the frequencies must add to a multiple of 5, 8, 16, and 20. Thus they must add to a multiple of the LCM, which is 80. Since there are only (about) 104 possible nights in the year, 80 must be the number of nights bookings were taken, and the number of nights 3 rooms were booked was $80 \times \frac{1}{5} = 16$.

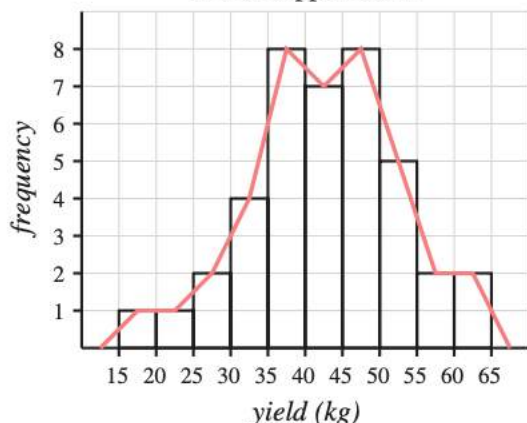
Exercise 15C

1a

lower	15	20	25	30	35	40	45	50	55	60
freq	1	1	2	4	8	7	8	5	2	2

b See the next answer.

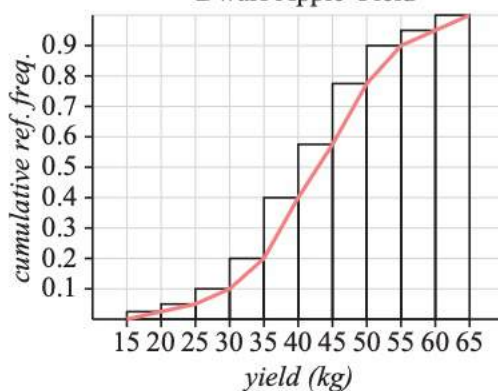
c Dwarf Apple Yield



e

lower	15	20	25	30	35	40	45	50	55	60
freq	1	1	2	4	8	7	8	5	2	2
cf	1	2	4	8	16	23	31	36	38	40
crf	0.025	0.05	0.1	0.2	0.4	0.575	0.775	0.9	0.95	1

f Dwarf Apple Yield



g About 43 kg

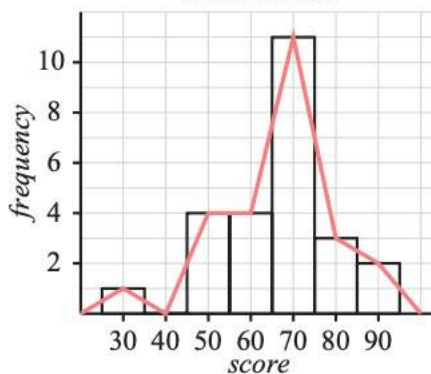
h i Trees falling below a 35 kg yield.

ii In terms of profit this may be a good idea, but consider also whether these represent a different variety of tree that sells well for other reasons, e.g. superior tasting fruit, fruiting early or late in the season, etc.

2a

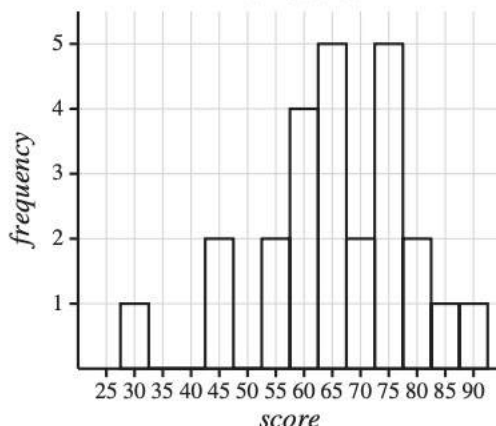
x	[25,35)	[35,45)	[45,55)	[55,65)	[65,75)	[75,85)	[85,95)
cc	30	40	50	60	70	80	90
f	1	0	4	4	11	3	2

b Exam Scores



c The grouped histogram is easier to interpret at a glance and shows a broader picture of those with similar results. However, data about individual results is lost.

d Exam Scores



3a X takes a real value, thus the probability that it is exactly 6 (rather than 6.01 or 5.993 or ...) is vanishingly small. Experimenters may record 6, because they tend to round off, but in the mathematical model X may take the full range of continuous real values. Thus $P(X = a) = 0$ for any value a .

b Also 0.2, as explained in the previous answer $P(X = 6) = 0$.

c i 0.25 ii 0.37 iii 0.45

4a 0.37

b $1 - 0.22 = 0.78$

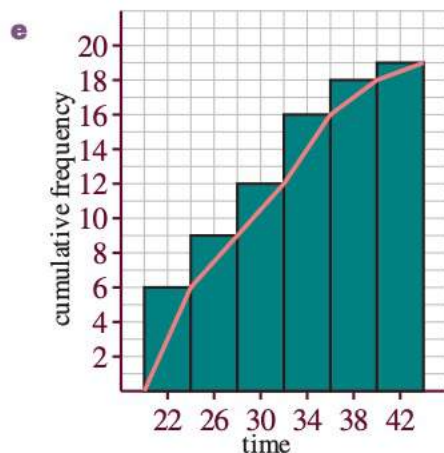
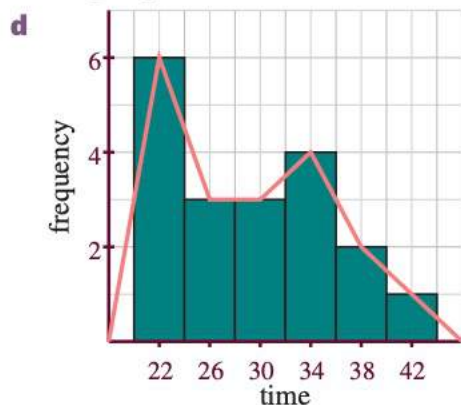
c i $0.79 - 0.12 = 0.67$

ii $1 - 0.57 = 0.43$

5a 29.5

class	20–24	24–28	28–32	32–36	36–40	40–44
centre	22	26	30	34	38	40
freq	6	3	3	4	2	1
cf	6	9	12	16	18	19

c 30. No, because information is lost when the data are grouped.



6a

x	152	154	155	157	158	159	162	163
f	1	2	1	1	2	3	2	3

x	164	165	166	168	170
f	2	2	3	1	1

b 162.5

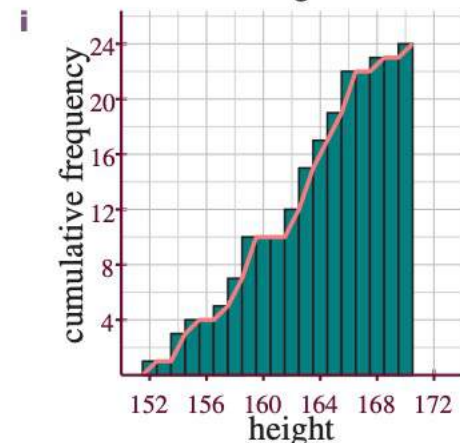
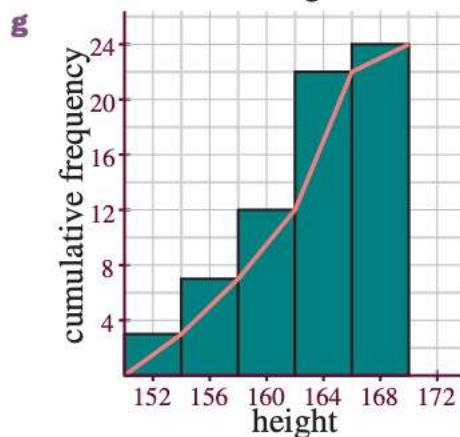
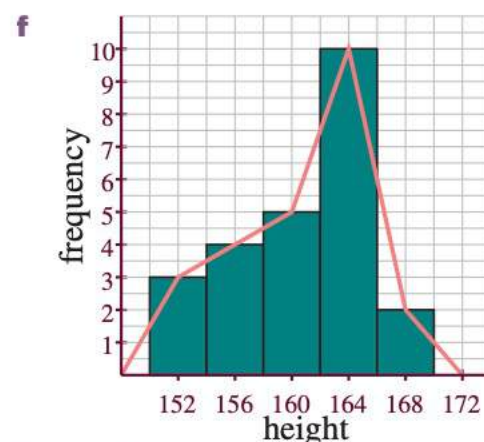
c Trends are less clear when the data are not grouped, because it is less visually clear that the data are falling in certain zones on the domain.

d

group	150–154	154–158	158–162	162–166
centre	152	156	160	164
freq	3	4	5	10

group	166–170
centre	168
freq	2

e 162



j The cumulative frequency polygon and cumulative frequency polygon are much less sensitive to the grouping process than the frequency histogram and frequency polygon. The graphs in parts **g** and **h** look very similar in shape.

7a $6 - 1 = 5$

b $Q_2 = 4, Q_1 = 3, Q_3 = 5$

c agrees

d The interval $[0.5, 1.5)$ has range 1.

e $6.5 - 0.5 = 6$

f $Q_2 = 4, Q_1 = 2.75, Q_3 = 5.25$

g The polygon on the histogram joins the endpoints of the columns, thus the correct answer is half-way between 3.5 and 4.5, viz 4. The value 0.48 represents the accumulation of scores up to 3.5, and 0.52 is the accumulation of scores up to 4.5

h The line $y - 0.52 = m(x - 4.5)$ with $m = 0.32$ gives $x = 5.21875$ when $y = 0.75$. The accuracy is not really justified - the original scores are given to 0 decimal places, possibly due to rounding. An answer like 5.25 suggests “quarter of a division past 5”, which is a reasonable interpretation without suggesting undue accuracy.

i 3.25

8a $0.105 \times 4 = 0.41$. Thus 41% of packages arrive in this time interval.

b $(0.045 + 0.025) \times 4 = 0.28$. Thus 28% of packages arrive in this time interval.

c 1. This is the total probability.

d This is the area under the polygon between 18 and 30 days. The area is a triangle, with area $\frac{1}{2} \times 12 \times .015 = 0.09$. The probability is 9%, or you could think of it as saying 9% of packages arrive then. The same answer could be obtained by finding the area under the last 2.5 rectangles.